

Nitrogen Atom Induced Contrast Effect on the Mechanofluorochromic Characteristics of Anthracene-Based Acceptor-Donor-Acceptor Fluorescent Molecules

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The mechanofluorochromic (MFC) characteristics of anthracene-based acceptor-donor-acceptor (A–D–A) fluorescent molecules are explored through a comprehensive investigation of their photophysical behaviors. Six 9,10-diheteroarylanthracene derivatives with varying acceptor groups (pyridin-4-yl, pyridin-3-yl, pyridin-2-yl, pyrimidin-5-yl, pyrazinyl and quinoxaliny) are synthesized and systematically characterized. The photophysical properties in both solution and solid-state are examined, revealing subtle yet significant influences of the spatial arrange-

ment and number of nitrogen atoms within the acceptor group on fluorescence emission. Single-crystal structures of these compounds provide insights into their steric configurations and intermolecular packing modes, offering valuable insights into the fundamental mechanisms that underlie the observed MFC properties. This study illuminates the intricate interplay between MFC properties and the refined molecular structure, thus presenting promising avenues for the design and advancement of novel MFC materials.

Introduction

Stimuli-responsive materials are a subclass of intelligent materials that manifest alterations in their physicochemical properties after being stimulated by external elements.^[1] Within this classification, there exists a category of materials that undergo notable modifications in fluorescence color upon exposure to external mechanical forces, such as shear, grinding, tension, or hydrostatic pressure, are denoted as mechanofluorochromic (MFC) materials.^[2] These substances inherently possess the capacity for dynamic and reversible fluorescence switching and can engender a visually striking array of chromatic manifestations simply through grinding or frictional manipulation of the compound. Furthermore, restoration of the manipulated material to its original state can be achieved through methodologies such as solvent fuming, heating, or other external stimuli. Owing to the distinctive properties exhibited by MFC materials, their potential applicability extends to the domains encompassing sensor technology,^[3] data storage,^[4] and the fabrication of anti-counterfeiting materials.^[5]

Nitrogen heterocycle-modified π -conjugates constitute a category of luminescent compounds that has been widely investigated. Researchers have successfully introduced nitrogen heterocycles into various luminescent moieties, such as tetraphenylethylene, anthracene, and naphthalene, thereby constructing a variety of compounds with outstanding luminescent

properties.^[6] Their studies have demonstrated that precise manipulation of the position of the nitrogen atom, while keeping the rest of the molecule unchanged, can significantly impact the optical properties of the molecule. Anthracene stands as a pivotal π -conjugated compound, renowned not only for remarkable luminescence properties but also for its ease of modifiability and superior electron-donating capacity, which has rendered it pervasive in the field of organic luminescent material synthesis.^[7] Despite the MFC characteristics of 9,10-distyrylanthracene derivatives, which have been extensively scrutinized,^[6b,d,8] a notable paucity of investigations pertains to 9,10-diheteroarylanthracene (DAA) compounds, thereby constraining the advancement of anthracene derivative-based MFC materials.^[9]

In this study, six acceptor-donor-acceptor (A–D–A) type 9,10-diheteroarylanthracene derivatives were designed and synthesized. The constituents of these compounds were meticulously chosen, with anthracene serving as the consistent donor group, and the acceptor groups varying as pyridin-4-yl (DAA-1), pyridin-3-yl (DAA-2), pyridin-2-yl (DAA-3), pyrimidin-5-yl (DAA-4), pyrazinyl (DAA-5), and quinoxaliny (DAA-6), as illustrated in Figure 1. By conducting comprehensive investigations into the photophysical characteristics of the six compounds in both solution and the solid state, our aim encompassed a meticulous examination of the impact stemming from microstructural adjustments of minute structural modulations within the acceptor group on the photophysical properties of the anthracene chromophore, with a specific focus on the modulation of MFC characteristics.

Results and Discussion

All compounds DAA-1 to DAA-6 were synthesized by Suzuki coupling reactions, and the detailed synthetic methods and

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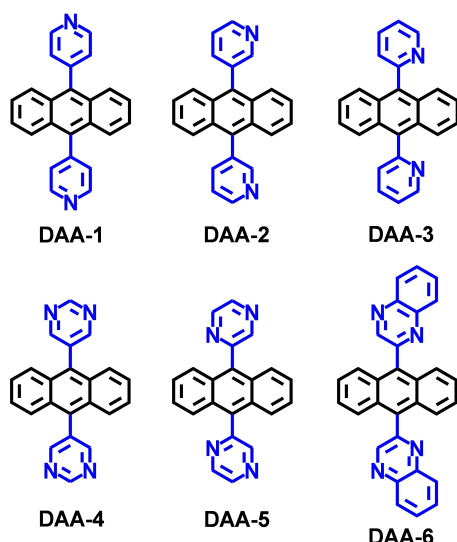


Figure 1. Molecular structures of six 9,10-diheteroaryl anthracene derivatives.

characterizations were shown in the Supporting Information (Scheme S1 and Figures S1–S23). Firstly, we conducted an investigation into the photophysical properties of these six compounds when dissolved in THF at a concentration of 10^{-5} M. As shown in Figure 2A, the UV-Vis spectra are depicted, revealing that **DAA-1** to **DAA-5** exhibit analogous absorption features. The absorption peak centered around 260 nm aligns with the aromatic heterocyclic absorption, concomitant with the characteristic anthracene absorption peak spanning 350 nm to 400 nm. **DAA-6** exhibited two absorption peaks at 254 nm and 317 nm, which can be attributed to the quinoxaline absorption. Noteworthy is the broader absorption in the 350–420 nm range, which signifies not only the inherent absorption traits of anthracene but also an intramolecular charge transfer transition from the anthracene donor moiety to the quinoxaline acceptor moiety. Figure 2B illustrates the fluorescence spectra, where the emission peak positions of **DAA-1**, **DAA-2**, and **DAA-4** are closely aligned in the range of 390 nm to 485 nm. The emission peak of **DAA-3** falls within the range of 393 nm to 502 nm. **DAA-5** exhibits an emission peak at 410 nm to 525 nm, while **DAA-6** displays an emission peak spanning from 413 nm

to 555 nm. The absolute emission quantum yields for **DAA-1** to **DAA-6** are 60 %, 61 %, 62 %, 63 %, 62 %, and 10 %, respectively.

Furthermore, we conducted a solvatochromism study on six compounds, and their absorption spectra remained unchanged in solvents with different polarities (Figure S24), indicating that the ground state is unaffected. The emission spectra of **DAA-1** to **DAA-5** also exhibited no significant changes, except for **DAA-6**, which exhibited variations in emission spectra in different solvents (Figure S25). The emission spectra of **DAA-6** red-shift with increasing solvent polarity, consistent with the charge transfer character of the emission band. The optimized geometries of six compounds were calculated by Density functional theory (DFT) at the B3LYP/6-31G (d, p) level with the Gaussian 09 software package. The frontier molecular orbitals further confirmed the charge transfer properties of **DAA-6**. The lowest unoccupied molecular orbital (LUMO) is mainly localized on the quinoxaline acceptors, while the highest occupied molecular orbital (HOMO) is mainly distributed on the anthracene segment (Figure S45), indicating that charge transfer can occur in **DAA-6**. In contrast, the HOMOs of **DAA-1** to **DAA-5** are mainly situated on the donor anthracene, with LUMOs distributed partially on anthracene as well as the acceptor groups (Figures S40–S44).

Subsequently, we further investigated the photophysical properties of **DAA-1** to **DAA-6** within the aggregated state. Fluorescence spectra of the six compounds were meticulously recorded in mixed solvent comprising varying proportions of tetrahydrofuran (THF) and water (H_2O), as shown in Figure 3. As the aqueous content in the solvent blend was gradually increased from 0 % to 90 %, distinct variations were observed across the six compounds. **DAA-1**, **DAA-2** and **DAA-3** exhibited a congruent pattern, fluorescence intensity were gradually enhanced with increasing the aqueous fraction in the solvent mixture, up to a threshold of 70 %. However, upon attaining a 90 % aqueous composition, a diminishing fluorescence intensity was observed, manifesting as a fluorescence quenching phenomenon, postulated to stem from the excessive aggregation of molecules (Figure 3A–3C). For **DAA-4**, a progressive enhancement in fluorescence intensity was discernible as the proportion of the H_2O component increased to 50 %, whereafter followed by a sequential decline between the 50 % and 90 % aqueous content (Figure 3D). In contrast, exclusively aggrega-

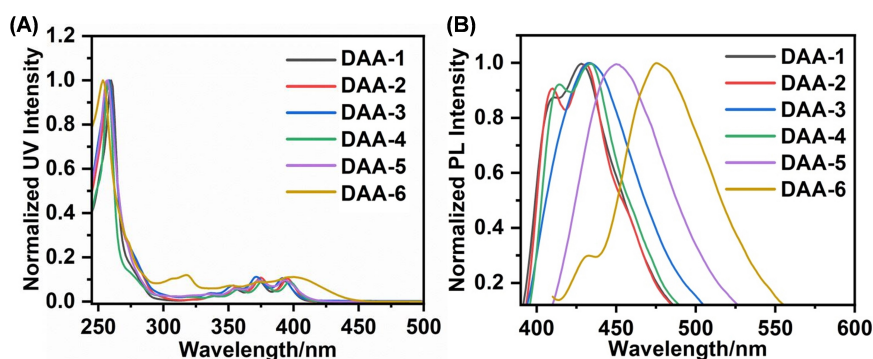


Figure 2. (A) UV-Vis and (B) fluorescence spectra of **DAA-1** to **DAA-6** in THF.

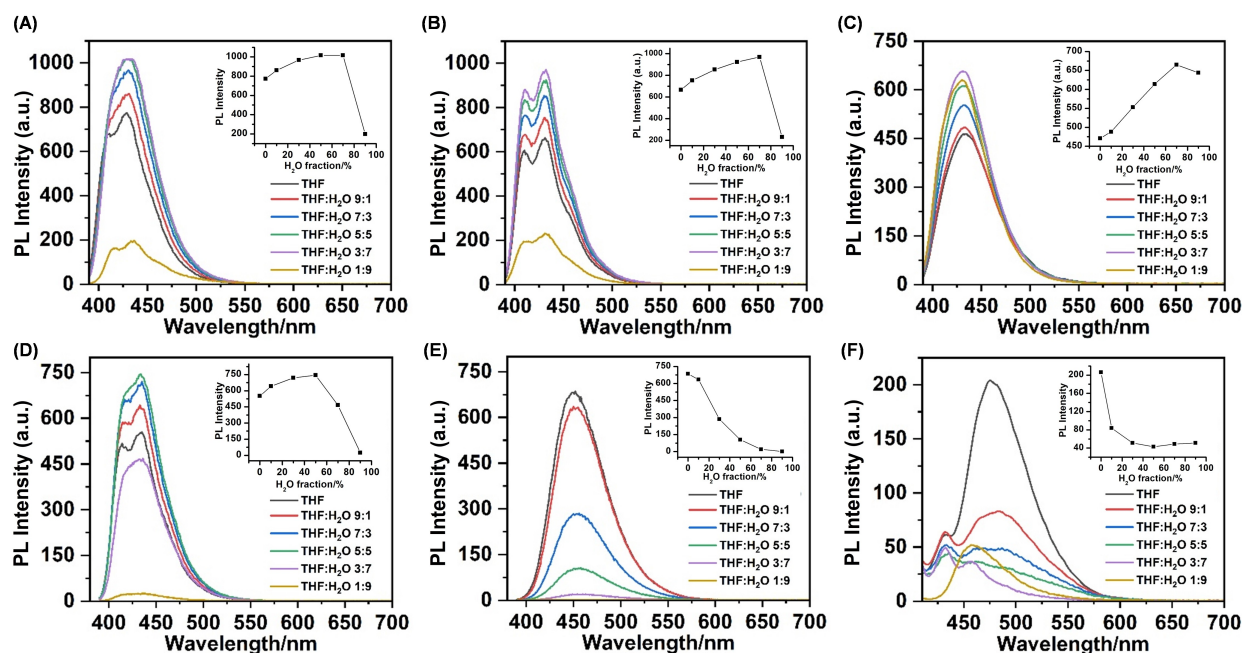


Figure 3. Fluorescence spectra of (A) DAA-1, (B) DAA-2, (C) DAA-3, (D) DAA-4, (E) DAA-5 and (F) DAA-6 in THF/H₂O mixtures. Inset: Plot of photoluminescence intensity versus H₂O fraction. ($c = 10^{-5}$ M, $\lambda_{\text{ex}} = 370$ nm for DAA-1 to DAA-5, $\lambda_{\text{ex}} = 400$ nm for DAA-6).

tion-caused quenching (ACQ) behavior was exhibited by DAA-5 and DAA-6 with increasing aqueous content (Figure 3E and 3F). We speculate that the gradual increase in initial fluorescence intensity for DAA-1 to DAA-4 with an increasing water content can be attributed to minor molecular aggregation, which restricts intramolecular motion, particularly involving single bond rotations between donor and acceptor units. The subsequent decrease in fluorescence intensity for DAA-1 to DAA-4, along with the appearance of ACQ phenomena in DAA-5 and DAA-6, can primarily be attributed to the enhanced π - π interactions between anthracene moieties, which suppress fluorescence emission. These findings emphasize that even minor variations within the acceptor groups can have discernible effects on the aggregated fluorescence properties of the final compounds.

After attaining a profound comprehension of the photophysical properties in solution, we proceeded to scrutinize the photophysical and MFC properties of six compounds in the solid-state. Each of these six compounds was thoroughly ground individually using an agate mortar, and subsequently subjected to pre- and post-grinding analyses. As demonstrated in Figure 4A and 4B, after grinding, DAA-1 with two 4-pyridyl acceptor groups, exhibited a discernible 10 nm red shift in its emission peak and reduced absolute emission quantum yield from $\Phi_F = 45\%$ (before grinding) to $\Phi_F = 32\%$ (after grinding). For both DAA-2 and DAA-3, featuring 3-pyridyl and 2-pyridyl acceptor moieties respectively, marginal redshifts were observed alongside pronounced peak broadening (Figure 4D and 4F). Absolute emission quantum yield measurements for DAA-2 displayed $\Phi_F = 12\%$ and $\Phi_F = 14\%$ pre- and post-grinding, while DAA-3 showcased values of $\Phi_F = 9\%$ and $\Phi_F = 15\%$. The alteration of fluorescence color from blue to cyan post-grinding

was a shared characteristic among these three compounds, indicative of inherent MFC traits (Figure 4A, 4C and 4E). Diverging from the pyridyl-anchored counterparts, DAA-4 to DAA-6 exhibited more significant MFC characteristics when the acceptor moiety was transmuted to a heteroaromatic ring featuring dual nitrogen atoms. For DAA-4 with two 5-pyrimidinyl acceptor groups, a significant broadening of the second emission peak were observed after grinding, accompanied by a reduction in absolute emission quantum yields from $\Phi_F = 9\%$ to $\Phi_F = 5\%$ (Figure 4G and 4H). Compare with DAA-4, DAA-5 modified by 2-pyrazinyl acceptor groups adjusted the position of two nitrogen atoms and underwent a sizeable 31 nm red shift post-grinding, concomitant with a discernible alteration in fluorescence color from blue to green. The absolute emission quantum yields of both pre- and post-grinding were 16% (Figure 4I and 4J). It is worth noting that DAA-6 features two 2-quinolinyl receptor groups, increasing the degree of molecular conjugation. After grinding, its primary emission peak exhibits the most pronounced redshift with an emission shift of 84 nm. Fluorescence color was significantly shifted from cyan to yellow, and absolute emission quantum yields before and after grinding were $\Phi_F = 5\%$ and $\Phi_F = 10\%$, respectively (Figure 4K and 4L). To summarize the experimental results, after grinding, the emission spectrum of DAA-1 exhibited a redshift of 10 nm, while DAA-2 to DAA-4 showed broadening of the emission peaks. DAA-5 and DAA-6 exhibited redshifts of 31 nm and 84 nm in their emission spectra, respectively. These findings underscore that the MFC properties of anthracene-based compounds are notably influenced by both the quantity (one or two) and spatial arrangement (position within the aromatic heterocycle) of nitrogen atoms within the acceptor group. It is worth noting that, owing to the

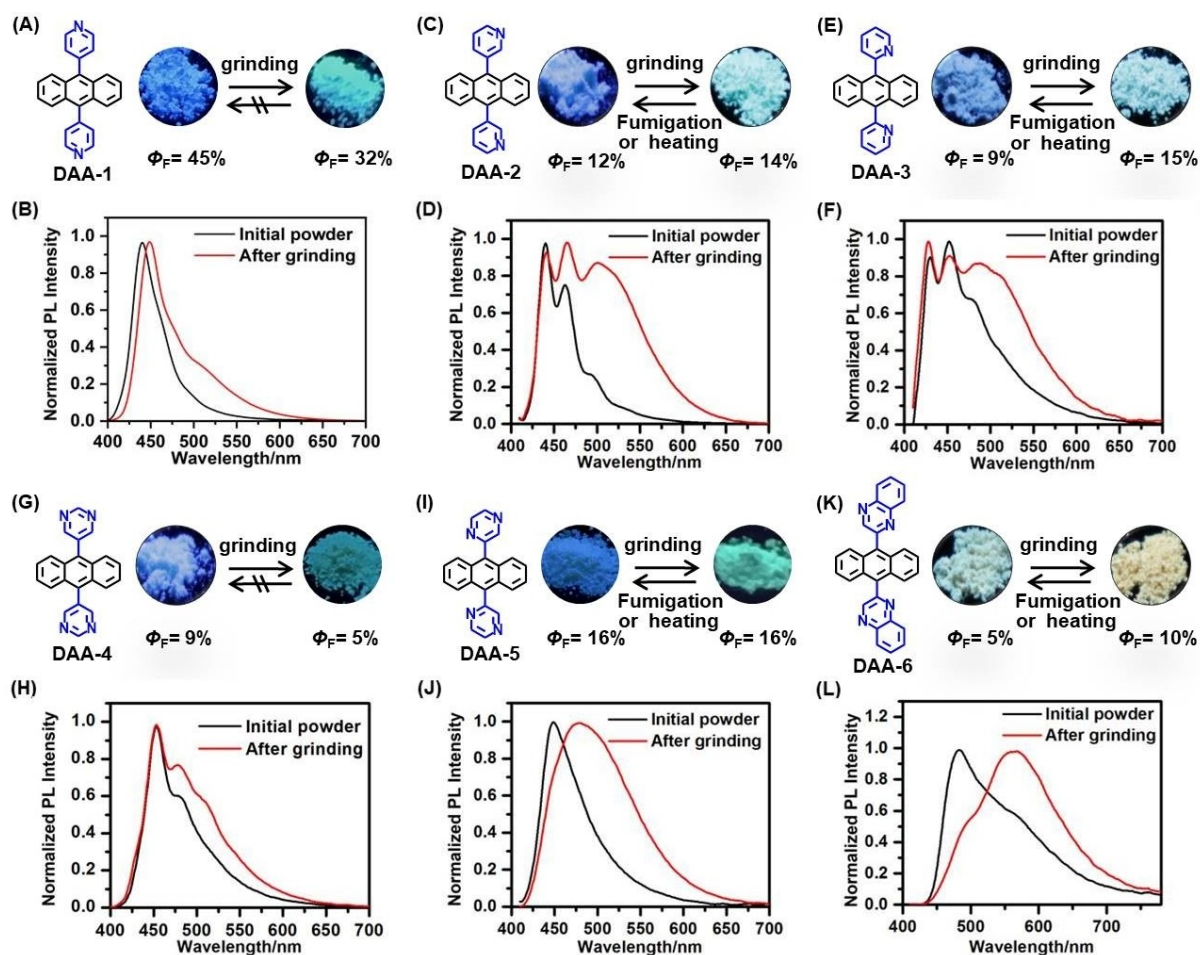


Figure 4. Solid-state fluorescence photographs and absolute emission quantum yields of (A) **DAA-1**, (C) **DAA-2**, (E) **DAA-3**, (G) **DAA-4**, (I) **DAA-5** and (K) **DAA-6**; Solid-state fluorescence spectra of (B) **DAA-1**, (D) **DAA-2**, (F) **DAA-3**, (H) **DAA-4**, (J) **DAA-5** and (L) **DAA-6** (λ_{ex} = 370 nm for **DAA-1** to **DAA-5**, λ_{ex} = 400 nm for **DAA-6**).

significant fluorescence changes observed in **DAA-6** after grinding, we explored its application in rewritable materials (Figure S29). Dispersion of **DAA-6** on filter paper, followed by pressing with a glass rod, revealed pronounced yellowing at the points of applied pressure. Additionally, the fluorescence color could be restored through CH_2Cl_2 fumigation.

During the course of powder X-ray diffraction (PXRD) analysis, it was observed that the initial solids of the six compounds displayed sharp and intense reflections, while the ground powders exhibited only slight peak broadening (Figure S30). This observation suggested the retention of an ordered microcrystalline structure in both pre- and post-grinding samples. Subsequently, we characterized the thermal behavior of these six compounds through thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC). The TGA analysis revealed that the decomposition temperatures for **DAA-1** to **DAA-5** were around 305 °C, while **DAA-6** exhibited higher stability with a decomposition temperature of 380 °C (Figure S31). Further, we conducted DSC analysis on the six compounds before and after grinding (Figure S32). For **DAA-1**, compared to the post-grinding sample, the original sample exhibited an additional endothermic peak at 150 °C. For **DAA-3**,

the original sample displayed an additional exothermic peak at 184 °C, while for **DAA-4**, the post-grinding sample showed an extra exothermic peak at 227 °C, indicating the presence of phase transitions within the partial microcrystalline structures of these three compounds after grinding. However, for **DAA-2**, **DAA-5**, and **DAA-6** in their two states (before and after grinding), we did not observe any additional endothermic or exothermic peaks.

Our investigation delved into the reversibility of these compounds. We systematically employed solvent fumigation and thermal treatment to ascertain whether the altered solid-state fluorescent colors of these compounds could be reinstated subsequent to the grinding process. Illustrated in Figure S28, the experimental results elucidate that the solid-state emission spectra of **DAA-2**, **DAA-3**, **DAA-5** and **DAA-6**, upon being subjected to CH_2Cl_2 fumigation or thermal treatment, exhibited a remarkable capacity for recovery, indicating their inherent nature of reversible fluorescence color change. In contrast, the emission spectra of ground **DAA-1** and **DAA-4** exhibited an absence of recovery following CH_2Cl_2 fumigation or heating, signifying an irreversible fluorescence color alteration. It is noteworthy that we extended our investigation to

other solvents, encompassing CHCl_3 , EtOH, THF, ethyl acetate and n-Hexane, however, these endeavors also proved incapable of restoring the fluorescence color of **DAA-1** and **DAA-4**, thus reinforcing the conclusion of irreversibility. Interestingly, compared with other four compounds, the nitrogen atom is located on the central axis of **DAA-1** and **DAA-4**. This distinction in behavior potentially stems from the permanent alteration in molecular arrangement of **DAA-1** and **DAA-4** post-grinding, thereby rendering their fluorescence properties impervious to remediation via external stimulus.

In an endeavor to elucidate the intricate interplay between the intrinsic MFC properties exhibited by these six compounds and their corresponding molecular structures, an in-depth investigation of their molecular aggregation states was pursued via single-crystal characterization. The controlled diffusion of isopropyl ether vapor into a chloroform solution containing the six compounds yielded the acquisition of crystals for each compound. The crystal structures of these compounds, along with their packing patterns within the unit cell, have been depicted in Figure 5, and crystallographic details can be found in the Supporting Information (Figures S33–S38 and Tables S1–

S6). It has come to our attention that each of the six compounds exhibits a sterically crowded structure marked by significant distortion. Specifically, among **DAA-1** to **DAA-6**, the dihedral angles (θ) of anthracene and acceptor motifs are 71.4° , 84.4° , 85.8° , 70.7° , 78.2° and 67.1° , respectively. In the crystal packing of six compounds, an abundance of weak intermolecular interactions is evident (Table 1). **DAA-1** exhibits 12 C–H... π interactions, 4 C–H...N interactions, and 4 π ... π interactions. **DAA-2** features 12 C–H... π interactions. **DAA-3** showcases 8 C–H... π interactions and 4 C–H...N interactions. **DAA-4** displays 10 C–H... π interactions, 6 C–H...N interactions, and 2 π ... π interactions. **DAA-5** exhibits the most intricate array of interactions, encompassing 16 C–H... π interactions, 12 C–H...N interactions, 8 π ... π interactions, 4 C...C interactions, 4 C...N interactions and 2 H...H interactions. Lastly, **DAA-6** presents 20 C–H... π interactions and 8 C–H...N interactions. Furthermore, we conducted an analysis of the crystal free volume fraction (FVF) through the Multiwfn program^[10] (Figure S39), resulting in FVF values of 28.43%, 27.96%, 30.69%, 29.89%, 22.56%, and 29.68% for **DAA-1** to **DAA-6**, respectively. The appropriate crystal void provides conditions for the transition in the intermolecular

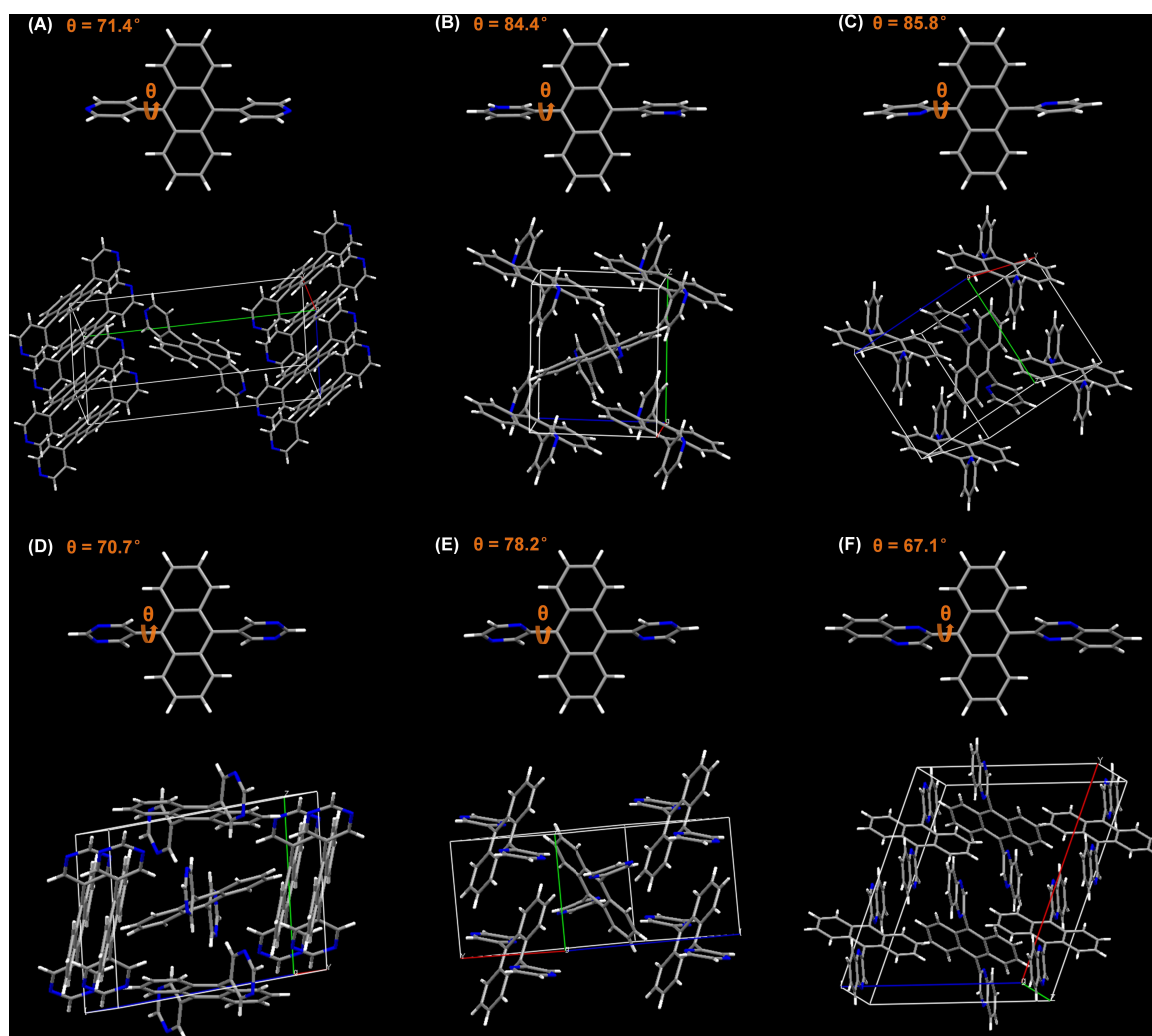


Figure 5. Single-crystal structures and packing modes of (A) **DAA-1**, (B) **DAA-2**, (C) **DAA-3**, (D) **DAA-4**, (E) **DAA-5** and (F) **DAA-6**.^[11]

Table 1. The list of various intermolecular interactions [Å] for DAA-1 to DAA-6.

	DAA-1 (20 interactions)	DAA-2 (12 interactions)	DAA-3 (12 interactions)	DAA-4 (18 interactions)	DAA-5 (46 interactions)	DAA-6 (28 interactions)
C-H... π	4*2.876	4*2.370	4*2.862	2*2.680	4*2.741	4*2.816
	4*2.897	4*2.894	4*2.898	2*2.748	4*2.810	4*2.823
	4*2.796	4*2.896		2*2.840	4*2.824	4*2.876
				2*2.848	4*2.877	4*2.880
				2*2.868		4*2.899
C-H...N	4*2.647		4*2.460	2*2.647	4*2.528	4*2.582
				2*2.732	4*2.571	4*2.673
				2*2.748	4*2.657	
π ... π	4*3.399			2*3.388	4*3.317 4*3.358	
C...C (not π ... π)					4*3.215	
C...N					4*3.050	
H...H					2*2.253	

stacking pattern following grinding. We postulate that the configuration of molecular distortion and various weak intermolecular interactions contribute to a loose intermolecular packing mode. Under the influence of external mechanical forces, the molecular configurations of the six compounds tend to favor relatively planar conformations, subsequently inducing intermolecular sliding. This phenomenon results in alterations in the stacking arrangement, ultimately leading to distinct redshifts in the fluorescence emission peaks of the six molecules and accompanying changes in fluorescence color.

By scrutinizing the single-crystal structures of six compounds, an enhanced comprehension of the MFC mechanism is attainable. The loose intermolecular packing mode, stemming from the distorted molecular architecture and multiple non-covalent interactions, affords ample space for the imposition of external stress conditions. Consequently, the molecules are poised to undergo distinctive conformational alterations when subjected to mechanical forces inducement, thereby engendering modulations in fluorescence properties. Furthermore, a salient observation pertains to the distinct acceptor groups, the number and arrangement of their nitrogen atoms exert a manifest influence upon the molecular packing, thereby significantly impacting the extent of red shift evident within the fluorescence emission peaks. Evidently, these discernments serve to unveil a profound connection between the MFC properties and the fine molecular structure of the compounds.

Conclusions

This study delves into the MFC properties of anthracene-based A–D–A type fluorescent compounds, focusing on the influence of the spatial arrangement and number of nitrogen atoms. Six 9,10-diheteroaryl anthracene derivatives with varied acceptor groups were synthesized and characterized. By systematically

analyzing their photophysical behavior in both solution and solid state, we have observed a substantial impact of the arrangement and number of nitrogen atoms within the acceptor molecule on its luminescent properties. By examining the single-crystal structures, the intricate steric configurations and intermolecular packing modes are revealed. The distorted molecular arrangements and multiple non-covalent interactions facilitate loose packing, enabling the accommodation of external mechanical forces and subsequent conformational changes. These adjustments ultimately lead to variations in fluorescence properties. This investigation establishes a nuanced relationship between MFC properties and the intricate molecular structure, providing valuable insights for the design and advancement of responsive luminescent materials.

Supporting Information

Synthetic details and characterizations of six compounds including NMR, MALDI-TOF-MS, spectral information, PXRD, TGA, DSC, X-ray crystallographic data and frontier molecular orbitals.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: mechanofluorochromic · anthracene · fluorescence · single crystal

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